

## 7 Cryptography

### 7.5 Modular Arithmetic Continued

Let's review what we have learned so far.

1. Start with definition of modular arithmetic.
2. Go over addition example, show that any number which has the same remainder can be used in a numbers place
3. Go over multiplication example, same thing
4. Go over exponentiaion example, same thing.

**Theorem 7.5.1** (*Modular Exponent Power Rule*) when  $n$  is a natural number, we have

$$(a^b \bmod n)^c \bmod n = (a^{bc} \bmod n) = (a^c \bmod n)^b \bmod n.$$

We also note, that when  $p$  is a prime number, it is always possible to find a ‘generator’, that is, a number  $g$  such that all the powers of  $g$  modulo  $p$  give all the possible values from 1 to  $p - 1$ .

**Example 7.5.2** Compute  $3^n \bmod 7$  for all values of  $n$ .

We are now ready to describe the key exchange algorithm.

1. Alice and Bob agree publically on values a prime  $p$  and generator  $g$ .
2. Alice picks some secret number  $a$ , while Bob picks some secret number  $b$ .
3. Alice computes  $A = g^a \bmod p$  and sends it to Bob.
4. Bob computes  $B = g^b \bmod p$  and sends it to Alice.
5. Alice computes  $B^a \bmod p$ , which is  $(g^b \bmod p)^a \bmod p$
6. Bob computes  $A^b \bmod p$ , which is  $(g^a \bmod p)^b \bmod p$

**Example 7.5.3** Suppose that Alice and Bob publicly choose the prime 17 and generator 3. What does the exchange look like, and where is their secret key?

**Example 7.5.4** Go through the key exchange algorithm with public values  $p = 17$  and generator  $g = 3$ .

**Example 7.5.5** Your turn! What does the key exchange algorithm look like with  $p = 19$  and  $g = 13$ ?